05-LinAlg-exc

Gauss and LU-factorization

1. To obtain the $\mathcal{O}(n^3)$ order of complexity of the forward row reduction algorithm of a $n \times n$-matrix, we used 49aa5f

$$s_n = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

the well-known identity for the sum of squares. Show it by induction.

2. Estimate the number of flops needed to forward row reduce of the linear system 766b47

$$A_{n\times n} X_{n\times m} = B_{n\times m} \colon (A|B) \sim \cdots \sim (U|B'),$$

so that $U_{n\times n} = [u_{ij}]$ is upper triangular, $u_{ij} = 0$ for $j > i$.

   (a) Find the #(flops) = (principal term)$(m, n) + \mathcal{O}(g(m, n))$.

   (b) Use (a) to decide what's more numerically expensive, to increase n (# of eqs/vars) or m (# of rhs)?